

- 1 One end of a light inextensible string of length a is attached to a fixed point O . The other end of the string is attached to a particle of mass m . The string is taut and makes an angle θ with the downward vertical through O , where $\cos \theta = \frac{2}{3}$. The particle moves in a horizontal circle with speed v .

Find v in terms of a and g .

[4]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook or legal stationery. There are no margins, text, or other markings on the page.